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* Gauss-Markov framework for linear model:

$$\begin{array}{c}
 Y = X \beta + \varepsilon \\
 \begin{array}{ccc}
 n \times 1 & n \times p & p \times 1 \\
 & \downarrow & \\
 & \text{non-random} &
 \end{array}
 \end{array}
 \quad \begin{array}{c}
 n \times 1 \\
 \hookrightarrow N_n(0, \sigma^2 I_n)
 \end{array}$$

Estimation for (β, σ^2)

$$\text{LSE: } \hat{\beta} = \arg \min_{\beta} \|Y - X\beta\|_2^2$$

$$\text{Est. for } \sigma^2: \hat{\sigma}^2 = \frac{\|Y - \hat{Y}\|^2}{n - \text{df}(\hat{Y})}$$

$$\hat{Y} = X\hat{\beta} \leftarrow \text{fitted values}$$

$$\text{df}(\hat{Y}) = \text{df}(HY) = P(H)$$

$$\uparrow \\ \text{Hat matrix} = P(X)$$

$$H = X(X^T X)^{-1} X^T \rightarrow \text{orthogonal projection matrix onto } \mathcal{C}(X)$$

What is $\hat{\beta}$?

$$\text{Normal equation: } X^T(Y - X\beta) = 0$$

$$\text{or } X^T X \beta = X^T Y$$

$$X^T Y \in \mathcal{C}(X^T) = \mathcal{C}(X^T X)$$

We know $Au = b$ solⁿ exists iff $b \in \mathcal{C}(A)$. $\therefore$ normal eqⁿ solⁿ exists.

For unique solⁿ: $P(X^T X) = p \leq n$, i.e., $X^T X$ is non-singular: $\hat{\beta} = (X^T X)^{-1} X^T Y$.

Theorem: The linear model, $Y \sim N_n(X\beta, \sigma^2 I_n)$ is identifiable iff $X^T X$ is non-singular.

Defⁿ: $P = \{P_0\}$ is identifiable iff $P_{\theta_1} = P_{\theta_2} \Leftrightarrow \theta_1 = \theta_2$

Proof:

Proof: $\Rightarrow N_n(X\beta_1, \sigma_1^2 I_n) = N_n(X\beta_2, \sigma_2^2 I_n) \Leftrightarrow ((\beta_1, \sigma_1^2) = (\beta_2, \sigma_2^2))$
iff $P(X) = p \Leftrightarrow X^T X$ is invertible

$\Leftarrow$ ~~Let~~ Let $X\beta_1 = X\beta_2$

Proof: $\Leftarrow$: Trivially done

$\Rightarrow$:

$\rightarrow$ Let $X^T X$ be singular

$$X^T X \hat{\beta} = X^T Y$$

$$X^T X (\hat{\beta} + z) = X^T Y \text{ if } z \in \mathcal{N}(X^T X)$$

Note: Generalized inverse: G is g -inverse of A if $AGA = A \Rightarrow$ for $Au = b$, Gb is a solⁿ.

$\hat{\beta}$ can be chosen to be $(X^T X)^- X^T Y$ for any g -inverse $(X^T X)^-$ of $X^T X$.

→ If A is symm., we can always find a symm. g-inverse of A .

$$A = P\Lambda P^T \text{ (spectral decomposition)}$$

$$PP^T = P^TP = I$$

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \text{ where } \lambda_i \text{'s are ordered eigenvalues of } A.$$

k non-zero ev's & $(n-k)$ 0 ev's.

Construct G s.t. $G = P\Delta P^T$ ($\Delta = \text{diag}(\delta_1, \dots, \delta_n)$)

$$AGA = A \Leftrightarrow P\Lambda P^T P\Delta P^T P\Lambda P^T = P\Lambda P^T$$

$$\Leftrightarrow \Lambda \Delta \Lambda = \Lambda$$

$$\Leftrightarrow \lambda_i^2 \delta_i = \lambda_i \quad \forall i=1, \dots, n$$

$$\Rightarrow \delta_i = \frac{1}{\lambda_i} \text{ if } \lambda_i \neq 0$$

$$\left\{ \begin{array}{l} \text{arbitrary if } \lambda_i = 0 \end{array} \right.$$

→ Moore-Penrose generalized inverse of A (symm.) ($A = P\Lambda P^T$) is of the form $A^+ = P\Delta P^T$ where $\Delta = \text{diag}(\delta_1, \dots, \delta_n)$ w/ $\delta_i = \frac{1}{\lambda_i}$ if $\lambda_i \neq 0$, 0 if $\lambda_i = 0$

$$\Lambda = \begin{pmatrix} \Lambda_1 & 0 \\ 0 & 0 \end{pmatrix} \text{ (}\Lambda_1 \text{ has only non-zero ev's)}$$

$$\Delta = \begin{pmatrix} \Lambda_1^{-1} & 0 \\ 0 & 0 \end{pmatrix}$$

This yields the smallest normed solution.

Note: "Estimable" means "can find unbiased estimator"

- i) What about estimation of $a^T \beta$?
- ii) Is $\hat{\beta}$ an unbiased estimator of β ?

(a) If $X^T X$ is non-random singular, then

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\Rightarrow E(\hat{\beta}) = (X^T X)^{-1} X^T E(Y) = (X^T X)^{-1} X^T X \beta = \beta$$

(b) If $X^T X$ is singular, consider a solⁿ $\hat{\beta}$ of the form $(X^T X)^- X^T Y$ for some g-inverse $(X^T X)^-$

$$E[(X^T X)^- X^T Y] = (X^T X)^- X^T E(Y) \\ = (X^T X)^- (X^T X) \beta$$

⊗ Notice! $\because X^T X$ isn't singular, β isn't actually using.

Result: $a^T \beta$ can be estimated unbiasedly if $a \in C(X^T) = R(X)$

Why?: $C(X^T) = C(X^T X) \rightarrow a = X^T X c$ for some c .

$$a^T \beta = c^T X^T X \beta$$

Let $\hat{\beta}$ be any solⁿ of $X^T X \beta = X^T Y$

$$a^T \hat{\beta} = c^T X^T X \hat{\beta} = c^T X^T Y$$

$$E(a^T \hat{\beta}) = c^T X^T E(Y) = c^T X^T X \beta = a^T \beta$$

Example: $Y_{ij} = \alpha_i + \beta_j + \varepsilon_{ij}$, $\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$, $i, j = 1, 2$

$$Y = \begin{pmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \end{pmatrix}, \beta = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \end{pmatrix}, X = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, P(X) = 3$$

$\therefore X$ doesn't have full rank, the linear model isn't identifiable, so we can't estimate β .

$a^T \beta$ is ~~only~~ estimable only when $a \in R(X)$

$$\Leftrightarrow a^T = c^T X = (c_1 + c_2, c_3 + c_4, c_1 + c_3, c_2 + c_4)$$

$$\therefore a \in R(X) \Leftrightarrow a_1 + a_2 = a_3 + a_4$$

So, now we can see that we can't estimate α_i or γ_i individually but we can estimate stuff like $\alpha_1 + \gamma_1$, $\gamma_1 - \gamma_2$, etc. (Just check ~~with~~ what $a^T \beta$ you're estimating and see if that a follows the above condition)

Result: ~~Sp~~ $a^T \beta$ is estimable, then $a^T \hat{\beta}$ is unique.

Why? Let $\hat{\beta}_1$ & $\hat{\beta}_2$ be 2 solⁿs of $X^T X \beta = X^T Y$

$$a^T \hat{\beta}_1 - a^T \hat{\beta}_2 = c^T X^T X (\hat{\beta}_1 - \hat{\beta}_2) = 0 \Leftrightarrow \hat{\beta}_1 = \hat{\beta}_2$$

Defⁿ: A linear unbiased estimator^(LUE) of $a^T \beta$ is an expression of the form $l^T Y$ where $l \in \mathbb{R}^n$.

Defⁿ: A LUE $l^T Y$ is said to be the best linear unbiased estimator (BLUE) for the parameter $a^T \beta$ if for any LUE $m^T Y$ of $a^T \beta$ we have:

$$\text{Var}(l^T Y) \leq \text{Var}(m^T Y)$$

Propⁿ: A linear parametric funcⁿ $a^T \beta$ is linearly estimable iff $a \in R(X)$

Why: $l^T Y$ is LUE for $a^T \beta \Leftrightarrow E(l^T Y) = a^T \beta \quad \forall \beta$

$$\Leftrightarrow l^T X \beta = a^T \beta \quad \forall \beta$$

$$\Leftrightarrow l^T X = a^T \Leftrightarrow a \in R(X)$$

Defⁿ A funcⁿ $C^T Y$ is called a linear zero funcⁿ (LZF) if $E(C^T Y) = 0$ for all β i.e., $C^T X \beta = 0 \forall \beta$
 $\Leftrightarrow X^T C = 0$

Propⁿ: A LUE $L^T Y$ is a BLUE iff $\text{Cov}(L^T Y, c^T Y) = 0$ for any LZF $c^T Y$.

Proof: If $m^T Y$ is a LUE, then $(m-l)^T Y$ is a LZF

$$\begin{aligned} \Leftarrow: \text{Var}(m^T Y) &= \text{Var}(l^T Y + (m-l)^T Y) \\ &= \text{Var}(l^T Y) + \text{Var}((m-l)^T Y) + 2 \underset{0}{\text{Cov}(l^T Y, (m-l)^T Y)} \end{aligned}$$

$$\Rightarrow \text{Var}(m^T Y) \geq \text{Var}(l^T Y)$$

$\Rightarrow l^T y$ is BLUE

$\Rightarrow$: Let $l^T y$ is BLUE

Let $C^T Y$ be LZF

For ~~λ any~~ arbitrary $\lambda \in \mathbb{R}$ $l^T y + \lambda c^T y$ is a LVE

$$\Rightarrow \text{Var}(l^T Y) \leq \text{Var}(l^T Y + \lambda c^T Y)$$

$$= \text{Var}(\ell^T Y) + \lambda^2 \text{Var}(c^T Y) + 2\lambda \text{Cov}(\ell^T Y, c^T Y)$$

$$\Rightarrow 0 \leq \lambda^2 \text{Var}(c^T Y) + 2\lambda \text{Cov}(l^T Y, c^T Y) \quad \forall \lambda \in \mathbb{R}$$

$\therefore \text{Var}(c^T Y) \geq 0$, if $\text{Cov}(l^T Y, c^T Y) \neq 0$, then

there will exist $\lambda \in \mathbb{R}$ s.t. $\nabla f(x) = \lambda \nabla g(x)$

$$\lambda^2 \text{Var}(c^T Y) + 2\lambda \text{Cov}(l^T Y, c^T Y) < 0 \text{ which will be a contradiction.}$$

$$\therefore \text{Cov}(l^T Y, c^T Y) = 0 \text{ if } l^T Y \text{ is BLUE. } \square$$